

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

Problem-Solving and Data Analysis

06

10 questions · ~15 min

Q1

PROBLEM-SOLVING AND DATA ANALYSIS

A fish swam a distance of 5,104 yards. How far did the fish swim, in miles?

1 mile = 1,760 yards

- A. 0.3
- B. 2.9
- C. 3,344
- D. 6,864

An analyst collected data on the price of a carton of grape tomatoes at 30 locations selected at random in Utah. The mean price of a carton of grape tomatoes in Utah was estimated to be \$ 4.23, with an associated margin of error of \$ 0.08. Which of the following is a plausible statement about the mean price of a carton of grape tomatoes for all locations that sell this product in Utah?

- A. It is between \$ 4.15 and \$ 4.31.
- B. It is either less than \$ 4.15 or greater than \$ 4.31.
- C. It is less than \$ 4.15.
- D. It is greater than \$ 4.31.

Q3

PROBLEM-SOLVING AND DATA ANALYSIS

Of 300,000 paper clips, 234,000 are size large. What percentage of the paper clips are size large?

- A. 22%
- B. 33%
- C. 66%
- D. 78%

	1 visit	2 or more visits	Total
Less than 40 years old	15	15	30
At least 40 years old	20	85	105
Total	35	100	135

The table summarizes customers who visited a car dealership in the last month by age and number of visits they made to the dealership. If a customer from the last month is selected at random, what is the probability that the selected customer is at least 40 years old?

- A. $\frac{30}{135}$
- B. $\frac{35}{135}$
- C. $\frac{100}{135}$
- D. $\frac{105}{135}$

Q5

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

73, 74, 75, 77, 79, 82, 84, 85, 91

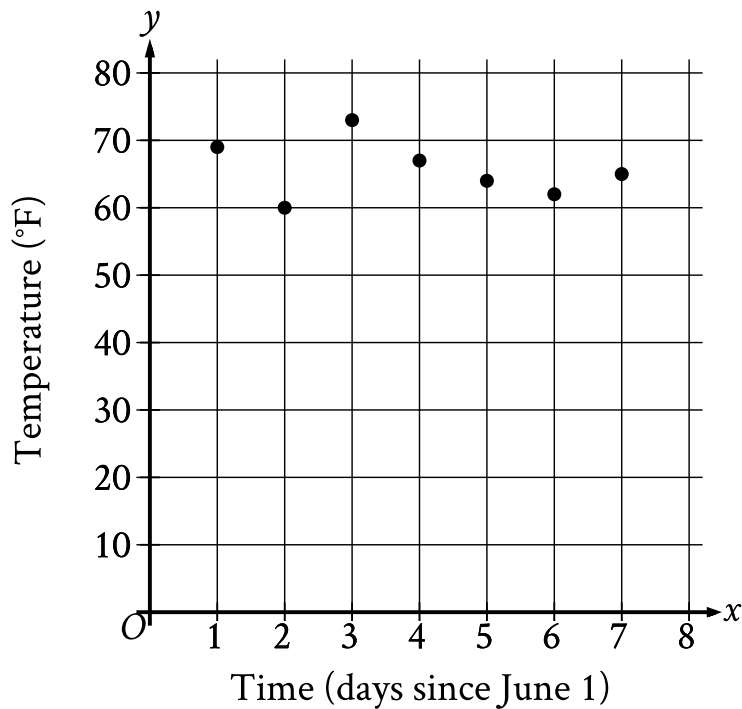
What is the median of the data shown?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The scatterplot shows the temperature y , in $^{\circ}\text{F}$, recorded by a meteorologist at various times x , in days since June 1.



- The scatterplot has 7 data points.
- The data points are in a linear pattern trending approximately horizontally.
- The data points have the following coordinates:
 - (1 comma 69)
 - (2 comma 60)
 - (3 comma 73)
 - (4 comma 67)
 - (5 comma 64)
 - (6 comma 62)
 - (7 comma 65)

During which of the following time periods did the greatest increase in recorded temperature take place?

- A. From $x = 6$ to $x = 7$
- B. From $x = 5$ to $x = 6$
- C. From $x = 2$ to $x = 3$
- D. From $x = 1$ to $x = 2$

Q7

PROBLEM-SOLVING AND DATA ANALYSIS

Makayla is planning an event in a 5,400-square-foot room. If there should be at least 8 square feet per person, what is the maximum number of people that could attend this event?

- A. 588
- B. 675
- C. 15,274
- D. 43,200

Q8

PROBLEM-SOLVING AND DATA ANALYSIS

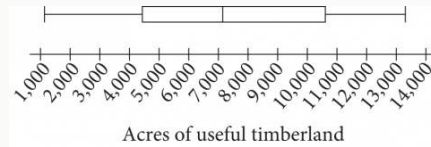
What is 20% of 440?

- A. 44
- B. 88
- C. 880
- D. 1,760

-11, -9, 26

A data set of three numbers is shown. If a number from this data set is selected at random, what is the probability of selecting a positive number?

- A. 0
- B. $\frac{1}{3}$
- C. $\frac{2}{3}$
- D. 1



The number of acres of useful timberland in 13 counties in California is summarized in the box plot above. Which of the following is closest to the median number of acres?

- A. 4,399
- B. 7,067
- C. 8,831
- D. 10,595